

Example 6: Create a Base RDD and Transform it (FINAL BOSS)

Here are the brief steps for writing the word counting program:

- Create a base RDD from Complete_Shakespeare.txt file.
- Use RDD transformation to create a long list of words from each element of the base RDD.
- Remove stop words from your data.
- Create pair RDD where each element is a pair tuple of ('w', 1)
- Group the elements of the pair RDD by key (word) and add up their values.
- Swap the keys (word) and values (counts) so that keys is count and value is the word.
- Finally, sort the RDD by descending order and print the 10 most frequent words and their frequencies.

Example 6: Step by Step Instructions

Step 1 Instructions:

- Create a RDD called baseRDD that reads lines from file_path.
- Transform the baseRDD into a long list of words and create a new splitRDD.
- Count the total number words in splitRDD.

Step 2 Instructions:

- Filter splitRDD, removing stop words listed in the stop_words variable.
- Create a pair RDD tuple containing the word (using the w iterator) and the number 1 from each word element in splitRDD.
- Get the count of the number of occurrences of each word (word frequency) in the pair RDD. Use a transformation which operates on key, value (k,v) pairs. Think carefully about which function to use here.

Step 3 Instructions:

- Print the first 10 words and their frequencies from the resultRDD RDD.
- Swap the keys and values in the resultRDD.
- Sort the keys according to descending order.
- Print the top 10 most frequent words and their frequencies from the sorted RDD.



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Big Data Processing

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Data Frames

What are DataFrames?

A DataFrame in PySpark is: “A distributed collection of data organized into named columns.”

It resembles a table in a relational database or a spreadsheet in which data is arranged in rows and columns. Each column can have a different data type, and DataFrames are highly optimized for parallel processing, making them suitable for handling large-scale datasets.

Key Features:

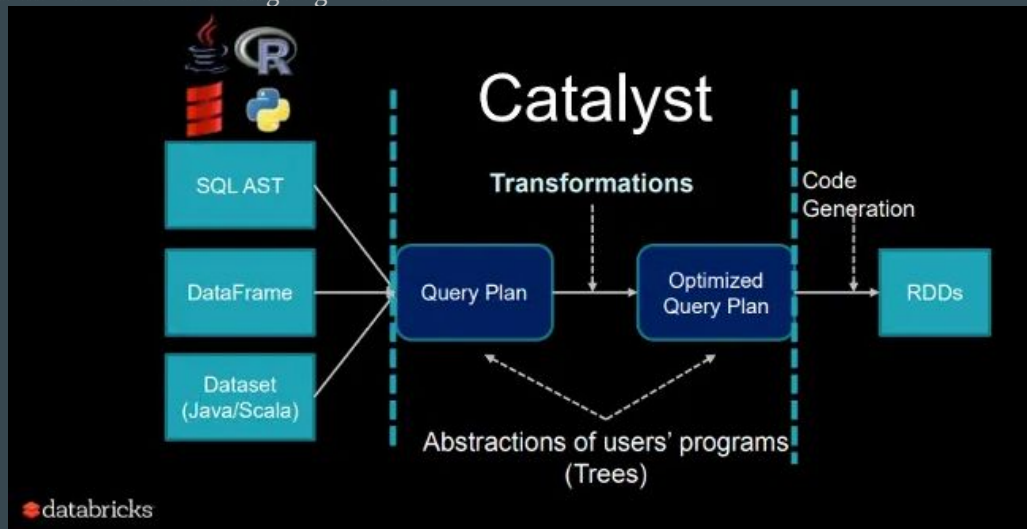
- ✓ Distributed & Scalable
- ✓ Schema-based (column names + types)
- ✓ Immutable (operations create new DataFrames)
- ✓ Lazy Evaluation (only executed on an action)
- ✓ Optimized with Catalyst Optimizer
- ✓ SQL-friendly

Real-world analogy:

Imagine you're analyzing sales data from 1,000 retail stores across the world.

In pandas, you'd load it into memory (might crash if it's 100 GB).

In PySpark DataFrames, the same analysis is split across a cluster → efficient, scalable, fault-tolerant.



Example 1: Create Dataframe from RDD

Create an RDD from the sample_list.

```
sample_list = [ ('Mona',20), ('Jennifer',34),('John',20), ('Jim',26)]
```

Create a PySpark DataFrame using the above RDD and schema.

Confirm the output as PySpark DataFrame. → print type.

```
1 # Create an RDD from the list
2 rdd = sc.parallelize(sample_list)
3
4 # Create a PySpark DataFrame
5 names_df = spark.createDataFrame(rdd, schema=['Name', 'Age'])
6
7 # Check the type of names_df
8 print("The type of names_df is", type(names_df))
```

Example 2: Loading CSV into Dataframe

We have people.csv now:

Create a DataFrame from file_path variable which is the path to the people.csv file.

Confirm the output as PySpark DataFrame.

```
1 # Create an DataFrame from file_path
2 people_df = spark.read.csv(file_path, header=True, inferSchema=True)
3
4 # Check the type of people_df
5 print("The type of people_df is", type(people_df))
```

Example 3: Inspecting Data

Note: Inspecting data is very crucial before performing analysis such as plotting, modeling, training etc.

Instructions:

- Print the first 10 observations in the people_df DataFrame.
- Count the number of rows in the people_df DataFrame.
- How many columns does people_df DataFrame have and what are their names?

```
1 # Print the first 10 observations
2 people_df.show(10)
3
4 # Count the number of rows
5 print("There are {} rows in the people_df DataFrame.".format(people_df.count()))
6
7 # Count the number of columns and print their names
8 print("There are {} columns in the people_df DataFrame and their names are {}".format(len(people_df.columns),
    people_df.columns))
```

Example 4: DataFrame subsetting and cleaning

After the data inspection, it is often necessary to clean the data which mainly involves subsetting, renaming the columns, removing duplicated rows etc.

Instructions:

- Select 'name', 'sex', and 'date of birth' columns from `people_df` and create `people_df_sub` DataFrame.
- Print the first 10 observations in the `people_df_sub` DataFrame.
- Remove duplicate entries from `people_df_sub` DataFrame and create `people_df_sub_nodup` DataFrame.
- How many rows are there before and after duplicates are removed?

```
1 # Select name, sex and date of birth columns
2 people_df_sub = people_df.select(['name', 'sex', 'date of birth'])
3
4 # Print the first 10 observations from people_df_sub
5 people_df_sub.show(10)
6
7 # Remove duplicate entries from people_df_sub
8 people_df_sub_nodup = people_df_sub.dropDuplicates()
9
10 # Count the number of rows
11 print("There were {} rows before removing duplicates, and {} rows after removing duplicates".format
      (people_df_sub.count(), people_df_sub_nodup.count()))
```

Example 5: Filtering DataFrame

In the previous exercise, you have subset the data using `select()` operator which is mainly used to subset the DataFrame column-wise. What if you want to subset the DataFrame based on a condition (for example, select all rows where the sex is Female). Let's see this example.

Instructions:

- Filter the `people_df` DataFrame to select all rows where sex is female into `people_df_female` DataFrame.
- Filter the `people_df` DataFrame to select all rows where sex is male into `people_df_male` DataFrame.
- Count the number of rows in `people_df_female` and `people_df_male` DataFrames.

```
1 # Filter people_df to select females
2 people_df_female = people_df.filter(people_df.sex == "female")
3
4 # Filter people_df to select males
5 people_df_male = people_df.filter(people_df.sex == "male")
6
7 # Count the number of rows
8 print("There are {} rows in the people_df_female DataFrame and {} rows in the people_df_male DataFrame".format(
    people_df_female.count(), people_df_male.count()))
```



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Stay Curious. Questions?

Thank you for your attention.
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